

UNIVERSITY OF RWANDA

Project Technical Report — Web Technology TECHNICAL ASSESSMENT 2026

Project Title: Twizere CyberNet
"Educate. Detect. Report. Repeat."

225067917
YEAR 1
19/07/2026

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List of Abbreviations

Abbreviation	Meaning
BNR	National Bank of Rwanda
RISA	Rwanda Information Society Authority
RURA	Rwanda Utilities Regulatory Authority
RIB	Rwanda Investigation Bureau
MFI	Microfinance Institution
SME	Small and Medium Enterprise
MoMo	Mobile Money
ERD	Entity Relationship Diagram
UI	User Interface
ICT	Information and Communication Technology

Chapter 1. Introduction

1.1. Historical Background of the Case Study

Rwanda is widely regarded as one of Africa's strongest digital transformation stories. Cashless payment systems are now common in Kigali's supermarkets, and even rural residents, such as a farmer in Rwamagana, can pay school fees through MTN Mobile Money without ever visiting a bank. This shift is driven by the government's Vision 2050 agenda, which pushes the country toward a fully digital economy. Mobile money transaction volumes have reached the trillions of Rwandan francs in recent years, and fintech startups continue to emerge across Kigali.

This rapid digitization, however, has outpaced the growth of cybersecurity awareness among both citizens and the institutions that serve them. Cybercriminals, both local and international, have identified this gap and increasingly target Rwanda's financial ecosystem through impersonation scams, fraudulent mobile money notifications, and phishing communications disguised as messages from trusted banks.

1.2. Problem Statement

Ordinary Rwandans regularly receive suspicious SMS messages pretending to be from institutions such as Bank of Kigali, along with fraudulent Mobile Money transfer notifications and phishing emails impersonating banks like Equity Bank or I&M Bank. Most victims only discover the fraud after their account has already been drained. This is not primarily a matter of carelessness — it reflects the fact that most citizens have never been taught what a cyberattack looks like in practice.

The problem is compounded inside financial institutions themselves. Banks, microfinance organizations, SACCOs, and fintech companies employ large numbers of staff who handle sensitive customer data daily, yet most have received no formal cybersecurity training. Studies across Africa show that over 90% of successful cyberattacks on financial institutions result from human error rather than sophisticated hacking. Further compounding this, Rwanda currently lacks a centralized system through which victims, banks, and regulators can jointly report, track, and analyze fraud patterns — meaning the same scam can victimize many people before it is detected and stopped.

1.3. Objectives of the Project

- To design and build a web-based platform that allows citizens to report and track financial fraud incidents in real time (FraudRadar RW).
- To provide a structured cybersecurity training module for employees of banks, SACCOs, MFIs, and fintech companies (FinCyber Academy).
- To enable users to submit suspicious messages, emails, or links for analysis and receive a phishing/fraud verdict (PhishBlock RW).
- To design a relational database (with ERD) that supports fraud reporting, training enrollment, phishing scans, and partner-institution collaboration.
- To model relationships between citizens, financial institutions, and external partners (regulators, telecoms, academia) so that fraud response is coordinated rather than fragmented.

1.4. Expected Outcomes

- A functional web platform where citizens can register, report suspected fraud, and view fraud trends in their area.
- A working training module where financial-sector employees can enroll in and complete cybersecurity courses.
- A phishing-scan feature that gives users an immediate verdict on suspicious content.
- A normalized relational database schema, derived from the project's ERD, capable of supporting all three pillars.
- A reduction, over time, in the isolation between how individual institutions currently handle fraud incidents.

1.5. Project Rationale (Importance to the Community)

Twizere — meaning "let's trust each other" in Kinyarwanda — reflects the philosophy behind the project: cybersecurity is not a problem any single citizen, bank, or regulator can solve alone. It requires a whole community reporting, learning, and protecting together. Rwanda already has the digital infrastructure, political will, and momentum needed; what has been missing is a platform that brings the community together around financial cybersecurity in a way that is accessible, intelligent, and specifically designed for the Rwandan context. By

connecting citizens, financial institutions, regulators, telecom operators, and academic partners on one platform, Twizere CyberNet aims to make Rwanda's digital financial space measurably safer.

Chapter 2. System Analysis and Design

2.1. System Analysis

2.1.1. Intended Users of the Project

These are the people who will directly use and benefit from Twizere CyberNet:

- Ordinary Rwandan citizens who use Mobile Money, online banking, and digital payment platforms.
- Employees of banks, SACCOs, MFIs, and fintech companies who need cybersecurity training.
- Small and medium business owners (SMEs) in Kigali and beyond conducting digital transactions.
- University students in ICT and finance programs using the platform as a learning and research tool.
- Independent journalists and financial bloggers verifying suspicious financial communications.

2.1.2. Intended System Partners

These are the organizations whose support, data, and authority make Twizere CyberNet credible and sustainable:

- National Bank of Rwanda (BNR) — primary financial regulator, providing authority and fraud data.
- Rwanda Information Society Authority (RISA) — drives digital transformation and cybersecurity policy.
- Rwanda Utilities Regulatory Authority (RURA) — helps flag and block fraudulent phone numbers.
- Rwanda Investigation Bureau (RIB) — acts on fraud reports submitted through FraudRadar RW.
- MTN Rwanda and Airtel Rwanda — track and flag Mobile Money fraud patterns.
- Bank of Kigali, Equity Bank, I&M Bank — primary institutional users of FinCyber Academy training.
- University of Rwanda (UR) and Carnegie Mellon University Africa — academic partners.
- The New Times and local media houses — spread awareness and amplify fraud alerts.

2.2. System Design

2.2.1. High-Level Architecture

The system follows a standard three-layer web architecture: a client layer handling presentation and input validation, a server/application layer handling business logic and authentication, and a database layer persisting all platform data. The three pillars (FraudRadar RW, FinCyber Academy, PhishBlock RW) sit within the application layer and share the same underlying database, allowing data such as user accounts and partner records to be reused across all three modules.

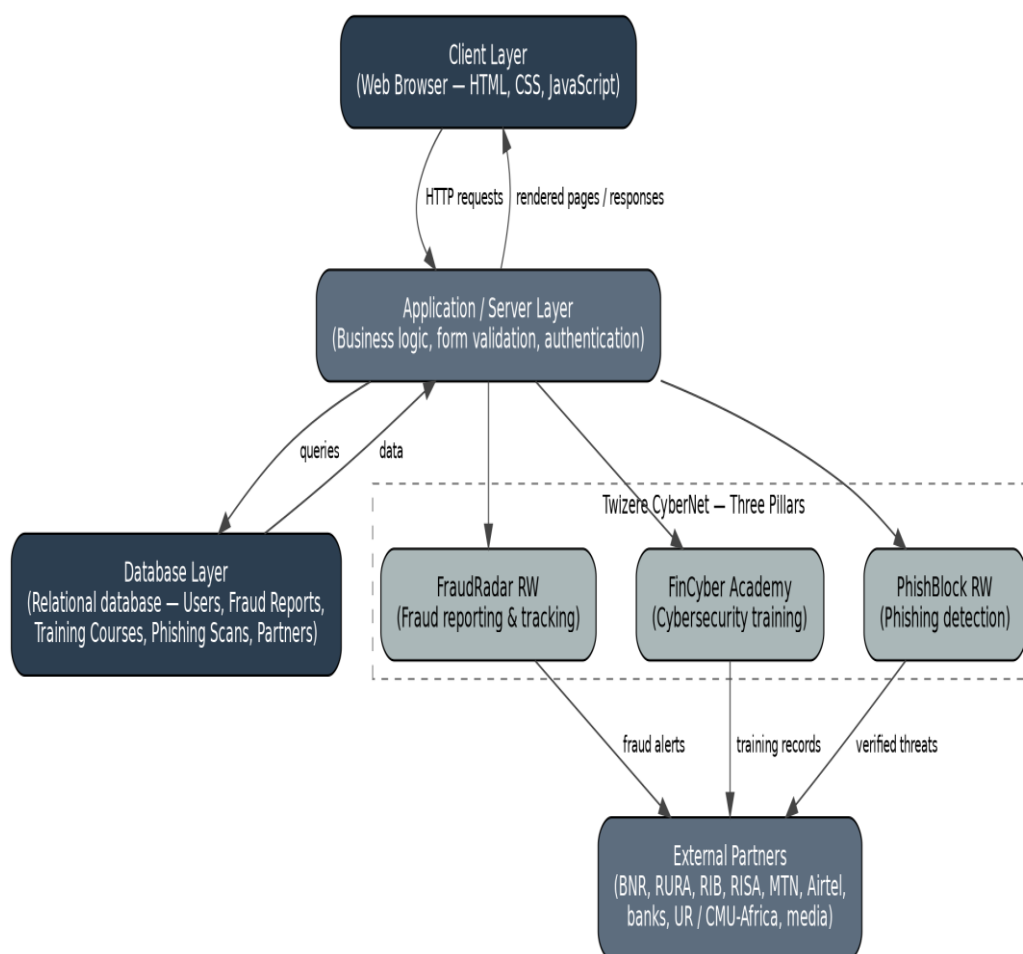


Figure 1: Figure 1: High-Level System Architecture

2.2.2. UI Design and Flowcharts

The diagram below shows the intended navigation flow: a visitor lands on the homepage, registers or logs in, and is then routed to a dashboard from which they can access any of the three pillars.

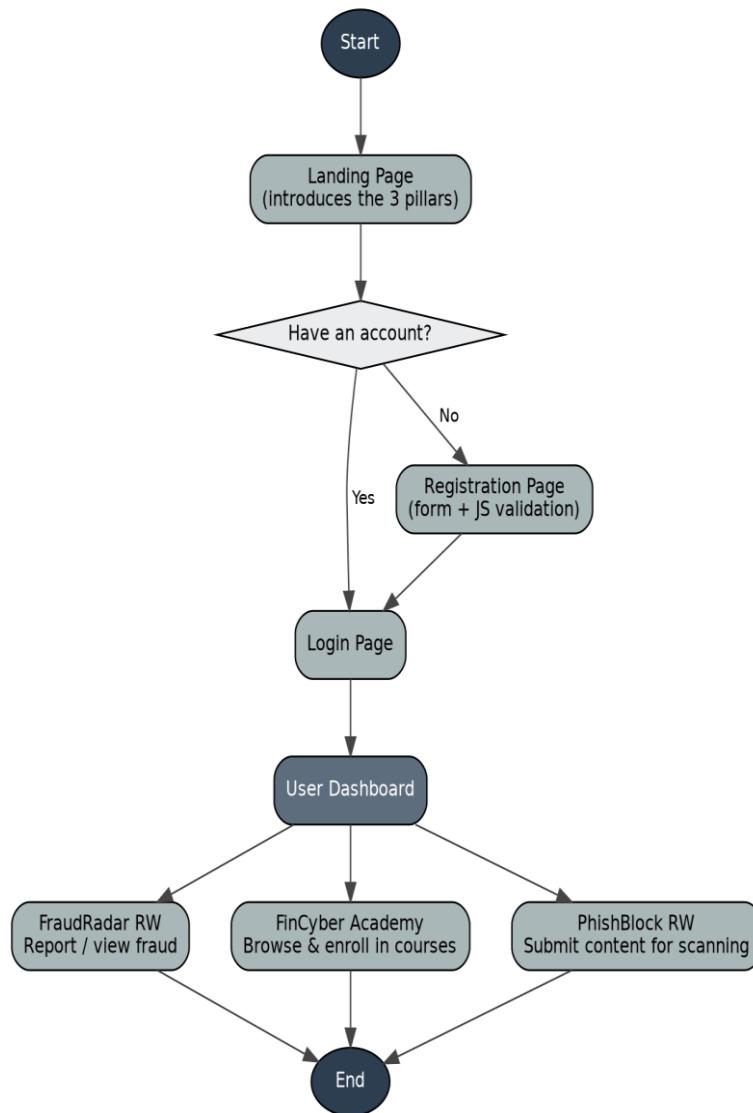


Figure 2: Figure 2: User Flow / Navigation Flowchart

2.2.3. Database Design with ERD and Relationships

The database is modeled around eleven entities. USER, INSTITUTION, and PARTNER are the core reference entities; FRAUD_REPORT, TRAINING_COURSE, and PHISHING_SCAN correspond to the three pillars; and EMPLOYMENT, ENROLLMENT, PARTNER_ACTION, and SCAN_ASSIST are associative (junction) entities that resolve the many-to-many relationships between users, institutions, courses, and partners.

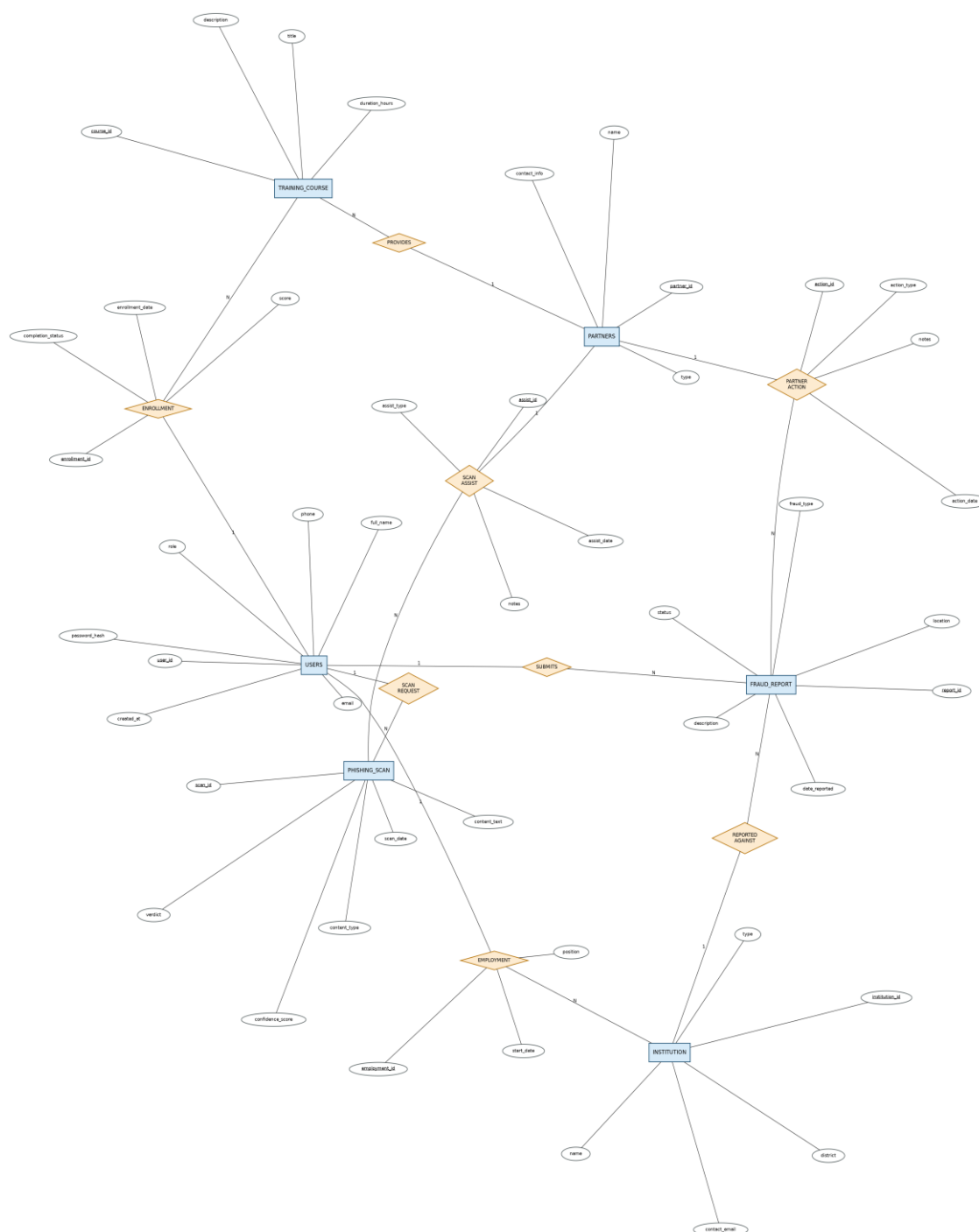


Figure 3: Figure 3: Entity Relationship Diagram (ERD)

Entity	Purpose
USER	Any registered person on the platform: citizen, employee, SME owner, student, or journalist.
INSTITUTION	Banks, SACCOs, MFIs, and fintech companies operating in Rwanda.
EMPLOYMENT	Links a user to the institution they work for (junction for USER–INSTITUTION).
FRAUD_REPORT	A fraud incident submitted through FraudRadar RW.
PARTNER	External bodies such as BNR, RISA, RURA, RIB, MTN, Airtel, UR, and CMU-Africa.
PARTNER_ACTION	What a partner did in response to a specific fraud report.
TRAINING_COURSE	A FinCyber Academy course, optionally provided by a partner.
ENROLLMENT	Links a user to the course(s) they are enrolled in (junction for USER–TRAINING COURSE).

PHISHING_SCAN	A message, email, or link submitted to PhishBlock RW for analysis.
SCAN_ASSIST	A partner's involvement in verifying or acting on a specific phishing scan (junction for PARTNER-PHISHING_SCAN).

Chapter 3. Implementation

3.1. Introduction

This chapter presents the pages implemented so far for Twizere CyberNet. The front end was built with HTML, CSS, and vanilla JavaScript, and covers all three pillars of the project: FraudRadar RW (fraud reporting), FinCyber Academy (training), and PhishBlock RW (phishing detection), linked together from a shared home page. At this stage the interface is fully built and interactive on the client side; it is not yet connected to the relational database described in Chapter 2, so submitted data is not yet persisted.

3.2. Screenshots and Discussion

3.2.1. Home Page (index.html)



Figure 4: Figure 4: Home Page

The home page introduces the platform under the tagline "Educate. Detect. Report. Repeat." and is structured as a series of stacked sections: an introduction to Rwanda's digital financial growth and the cyber threats that come with it, a description of the FraudRadar RW solution, a list of the platform's three core services, a list of who can use the platform (citizens, bank employees, SME owners, journalists, and students), and a list of key partners (BNR, RISA, RURA, RIB, MTN, Airtel, and commercial banks). A dynamic welcome message ("Welcome to Twizere CyberNet! Stay safe online.") is injected via JavaScript on page load. The page closes with a "Take Action" section linking out to the three functional pages: Report, Learn, and Detect. Layout is built with HTML tables rather than CSS Grid or Flexbox, styled through a shared style.css with a translucent pink table background over a fixed full-page background image.

3.2.2. Report Cyber Fraud Page — FraudRadar RW (report.html)

Report Cyber Fraud
Help us fight cybercrime in Rwanda

Full name:

Email address:

phone number:

Type of issue:
phishing ▾

DESCRIPTION:

Date

[Back To Home](#)

Figure 5: Figure 5: Report Cyber Fraud Page (FraudRadar RW)

This page implements the fraud-reporting form behind FraudRadar RW. It collects the reporter's full name, email address, and phone number; a dropdown for the type of issue (phishing, fake SMS, fraud, other); a free-text description text area; and a date field. These fields map directly onto the USER and FRAUD_REPORT entities from the database design in Chapter 2 — full name, email, and phone belong to USER, while the issue type, description, and date belong to FRAUD_REPORT. On clicking "Report Now", a JavaScript handler currently displays a confirmation message ("Your cyber fraud report has been submitted successfully") directly on the page; the form does not yet perform field validation or send data to a server, so the next implementation step is to validate required fields and wire the form to insert a row into FRAUD_REPORT (and, indirectly, USER).

3.2.3. FinCyber Academy Page (lesson.html)

FinCyber Academy
Learn How To Stay Safe Online
Cybersecurity education for every Rwandan

Why Cybersecurity Education Matters
Many cyber attacks succeed because people do not know how online scams work. FinCyber Academy helps citizens, students, businesses and bank workers understand common cyber threats and how to avoid them.

Lessons Available

Phishing Emails - How fake emails look - Dangerous links and attachments - How hackers steal passwords - How to verify official emails	Fake SMS Messages - Detect fake Mobile Money alerts - Avoid sharing secret PINs - Recognize suspicious phone numbers - How to report fraud messages
Password Security - Create strong passwords - Avoid easy passwords - Use different passwords - Protect bank accounts online	Safe Internet Usage - Avoid unsafe websites - Understand online privacy - Protect personal information - Use secure internet connections

Who Can Learn Here?

- Students
- Bank workers
- Business owners
- Ordinary citizens
- Fintech employees
- Journalists and bloggers

Our Goal
To create a cyber-aware Rwanda where citizens can safely use digital financial services without fear of scams and fraud.

[Back To Home](#)

Figure 6: Figure 6: FinCyber Academy Page

The FinCyber Academy page presents the training pillar of the platform. It explains why cybersecurity education matters, then lists four lesson categories in a grid: Phishing Emails, Fake SMS Messages, Password Security, and Safe Internet Usage, each with its own set of bullet-point topics. Further sections list who can learn on the platform (students, bank workers, business owners, citizens, fintech employees, journalists) and state the platform's overall goal. A "Start Learning" button triggers a JavaScript message confirming the user has begun a lesson. In terms of the database design, each of the four lesson categories corresponds conceptually to a row in `TRAINING_COURSE`, and the "Start Learning" action is the point at which an `ENROLLMENT` record would be created once the page is connected to the backend.

3.2.4. PhishBlock Page (detector.html)

Figure 7: Figure 7: PhishBlock Page

The PhishBlock page implements the phishing-detection pillar. Users paste a suspicious message or link into a text area, select where they received it (Email, SMS, WhatsApp, Social Media, Other), and indicate whether the message asked for money or passwords. Clicking "Scan Message" runs a client-side JavaScript keyword check: if the pasted text contains words such as "password", "pin", "bank", "click", "verify", "urgent", "account", or "money", the page displays a warning that the message looks suspicious; otherwise it reports that no obvious phishing keywords were found. This maps onto the `PHISHING_SCAN` entity — the pasted text becomes `content_text`, the source dropdown becomes `content_type`, and the keyword-match outcome becomes `verdict`. The page also lists common warning signs of phishing and closes with general safety advice. As with the other two pillars, the scan logic currently runs entirely in the browser and does not yet write a `PHISHING_SCAN` row to the database, and the keyword list is a simple placeholder for the more thorough detection logic the final system will need.

3.3. Shared Styling and Interactivity

All four pages share a single `style.css` file, which sets a fixed full-page background image, centers page content in tables at 80% width, and gives every table a translucent pink background for readability against the image. All interactivity — the home page welcome message, the report confirmation, the phishing keyword scan, and the lesson start message — is handled by a single `javascript.js` file using plain DOM manipulation (`getElementById` and `onclick` handlers), with no external libraries or frameworks.

Chapter 4. Conclusion and Recommendation

At this stage of the project, the front-end interface for all three pillars of Twizere CyberNet is complete and functional in the browser: the home page, the FraudRadar RW reporting form, the FinCyber Academy lesson page, and the PhishBlock RW detection page are all built, styled, and interactive. The database has been fully designed but not implemented, with an eleven-entity ERD covering users, institutions, partners, and all three pillars, along with the associative entities needed to resolve its many-to-many relationships.

For now I can say we completed 40% of our project.

The main piece of work remaining is backend integration: implementing the relational database from the ERD, connecting each form (report.html, lesson.html, detector.html) to that database so submissions are actually stored, adding user authentication (registration and login), and replacing the client-side keyword check in PhishBlock RW with more thorough phishing-detection logic.

Recommendations

- To the lecturer, on technicalities: the current submission demonstrates front-end design, JavaScript interactivity, and full database/ERD planning; backend implementation (server-side code and live database) is still in progress and is the next milestone.
- To the school, on time constraints: the time line allotted for this module and access to computers made it difficult to complete well this project, longer time and a phased submission structure with separate check points in design and implementation would allow us students to deliver more fully working systems.
- To the partners (owners of the business targeted), about the app developed: once connected to a live database, the platform will allow partner institutions to receive fraud reports and phishing alerts directly rather than through informal channels.
- To the user, about training: the FinCyber Academy lessons are most effective when paired with the practical habit of checking messages through PhishBlock RW before acting on them.
- To the beneficiaries (customers/students), about the new technological tool: the platform is designed to be used regularly rather than only after a suspected fraud attempt, since the training content is intended to build lasting awareness.

Appendix

Link to the GitHub repository (with account and password, if the repository is private) of the deployed project:

<https://irakozeaphia.github.io/twizere-cybernet/> TO SEE THE IMPLEMENTED WEBSITE

<https://github.com/irakozeaphia/twizere-cybernet> TO SEE THE WEBPAGES(html,css,javascript)